

Robustness Analysis of Real and Synthetic Networks

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Abstract

This project examines the structural robustness of complex networks, building on the work of Albert et al. [4]. Their study showed that a network's topology strongly influences how it reacts to node removal. Scale-free networks, such as the Internet and the World Wide Web, are highly tolerant to random failures but very vulnerable to targeted attacks on high-degree nodes (the so-called hubs). In contrast, networks displaying degree distributions with exponential tails, behave differently: they are more sensitive (with respect to the power law tailed scale-free networks) to random errors but more resistant to hubs removal. In this work, we extend the analysis in two main directions. First, we consider a broader set of networks, including both synthetic models and real-world networks. Second, we introduce and evaluate an attack strategy based on *betweenness centrality*. Following the results of Holme et al. [6], there are strong reasons to expect that betweenness-based attacks, especially when recalculated dynamically during the removal process, are among the most destructive strategies for many network topologies. Our findings, however, show that such superior effectiveness depends greatly on the characteristics of the specific network and is, at times, falsified.

1 Introduction

The study of network robustness has attracted significant attention over the past few decades because of its importance in understanding and predicting the effects of failures in complex systems. Many real-world systems can be naturally represented as networks (including the Internet, power grids and transportation systems) and, due to their interconnected structure, local failures may affect global properties of the system and severely reduce its functionality. Consequently, assessing network robustness is essential for evaluating the resilience of these systems against random failure and targeted attacks.

In this project, we investigate several real-world networks, including power grids, transportation networks, social networks, and a web graph, together with a number of synthetic network models: Erdős-Rényi, Barabási-Albert, Stochastic Block Model, and Watts-Strogatz networks. We begin by characterizing their global structure through a set of standard topological measures, with the goal of identifying the main structural features of each network. We then evaluate their robustness by simulating three different node-removal strategies: random attacks, highest-degree attacks, and highest-betweenness attacks. Throughout this work, the term attack refers to the removal of a node together with all its incident edges.

To monitor the response of the networks during the attack process, we track two key quantities: the normalized size of the Giant Connected Component (GCC), denoted by S , and the average size of the non-GCC connected components, denoted by $\langle s \rangle$. These metrics provide complementary information about the fragmentation dynamics of the network.

We use the results to assess whether or not the network structure we expected from the characterization, is compatible with the response to this robustness analysis. Hence we obtain both information on the networks structures and on their robustness.

2 Theoretical and Methodological Framework

In this section we will describe the measures used for the characterization (2.1) of the networks with a focus on the centralities (2.2). We also describe the attack strategies (2.3) as well as the models used for the synthetic networks (2.4)

2.1 Global Network Properties

To characterize the global structure of the networks, we consider a set of standard complex network metrics capturing size, connectivity, and topology. Network size is quantified through the **number** of nodes N and **edges** L , while the **density** and the **average degree** are used to learn about the connectivity of the network, since they reveal how sparse the networks are and the typical number of connections per node. For directed networks, it is necessary to distinguish between in-degree (k_{in}) and out-degree (k_{out}), which can differ from one another. We also evaluate the **degree variance** to measure the heterogeneity of the degree distribution.

The **global clustering coefficient**, measured as the ratio between closed triangles and all connected triplets of nodes, quantifies the global tendency of the network to be highly interconnected. High clustering introduces redundancy in the links, increasing the overall robustness of the network.

The **average shortest path length** represents instead the mean length of the shortest paths between randomly selected pairs of nodes and indicates the velocity at which information can travel across the network.

The **GCC** identifies the largest connected component of the graph. In directed networks, two distinct definitions are used: the weak GCC, where the network is treated as undirected and link directions are ignored, and the strong GCC, where nodes must be mutually reachable while respecting edge directions. For directed graphs, metrics based on reachability, such as the average shortest path length, are computed on the strong GCC, ensuring the existence of directed paths between all node pairs.

Degree assortativity indicates whether nodes tend to connect to others with similar connectivity levels (i.e. similar degree); negative values indicate dissortativity.

Finally, the community structure is analyzed through the **number of detected communities** and the corresponding modularity, which quantifies how well the network can be partitioned into highly connected groups that are weakly connected to one another. **Modularity** is computed using the Leiden algorithm via:

```
leidenalg.find_partition(graph, ModularityVertexPartition)
```

implemented in the **igraph** library.

The average shortest path length and the clustering coefficient are computed exactly or through sampling, depending on the size of the network.

2.2 Centrality Measures

To consider local measures on nodes and support the robustness analysis, we examine three centrality metrics: degree centrality, PageRank, and betweenness centrality. Each metric represents a different notion of influence, and for this reason we evaluate the Spearman rank correlation between the corresponding centrality scores. High correlations indicate that the measures agree on the importance of nodes, while negative correlations indicate an inverse relationship. Low correlations around zero instead suggest that the metrics quantify structural roles that are independent from one another and may therefore lead to different outcomes under targeted node removal strategies.

Degree Centrality

Degree centrality measures the total number of edges attached to a node. In an undirected network,

$$C_D(i) = k_i,$$

where k_i is the degree of node i .

For directed networks instead:

$$C_D^{\text{in}}(i) = k_i^{\text{in}}, \quad C_D^{\text{out}}(i) = k_i^{\text{out}}.$$

Degree centrality provides a local measure of importance and identifies hubs with many direct connections. From a robustness perspective, high-degree nodes are often critical because their removal destroys a large number of links simultaneously and can significantly reduce network connectivity.

PageRank Centrality

PageRank considers not only the number of connections but also their quality. We compute it using the function `igraph.Graph.pagerank(graph)`. The PageRank score of node i is defined recursively as

$$PR(i) = \frac{1-d}{N} + d \sum_{j \in \Gamma^-(i)} \frac{PR(j)}{k_j^{\text{out}}},$$

where N is the number of nodes, $\Gamma^-(i)$ denotes the set of nodes pointing to i and d is the damping factor (we will use its typical value $d = 0.85$). The term $(1-d)/N$ represents the PageRank contribution for nodes with no incoming links.

Betweenness Centrality

Betweenness centrality measures how often a node appears on shortest paths between other pairs of nodes. It is defined as

$$C_B(i) = \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}},$$

where σ_{st} is the number of shortest paths between nodes s and t , and $\sigma_{st}(i)$ is the number of those paths passing through node i .

Nodes with high betweenness are traversed by many shortest path, hence they may act as bridges between different regions of the network. Their removal can significantly affect the network topology, often causing fragmentation and reducing overall connectivity.

To ensure computational scalability on large networks, we make use of an approximation algorithm for betweenness centrality. In particular, we use NetworKit's **ApproxBetweenness** algorithm, which computes betweenness within a probabilistic error controlled by the parameters ϵ and δ [7]. Here, ϵ sets the maximum allowed error in the estimate, while δ specifies the probability that the error exceeds ϵ .

Spearman Rank Correlation

To compare the centrality measures, we use the Spearman rank correlation coefficient. Given two centralities X and Y , the coefficient is defined as

$$\rho = 1 - \frac{6 \sum_{i=1}^N d_i^2}{N(N^2 - 1)},$$

where d_i is the difference between the values of X and Y assigned to node i and N is the number of nodes.

Spearman correlation evaluates how consistently two rankings order the nodes, rather than the magnitude of their centrality scores. If correlation is high, this means that different notions of node importance identify the same vulnerable nodes, while low correlation indicates that each measure highlights distinct structural characteristics.

2.3 Attack Strategies

To evaluate network robustness, we simulate node failures by removing, at each step, the same number of nodes ΔN (which depends on the size of the network) together with all their incident edges. This

corresponds to removing a part of the network and all its connections to the rest of the system. The value of ΔN depends on the step n and is determined by the formula

$$N_n = n \times dq \times N_0$$

where N_n indicates the number of nodes at after n removal rounds, $N_0 = N$ is the number of nodes of the original network and we set $dq = 0.005$.

For each network, we consider three attack strategies: random attack, recalculated highest-degree attack, and recalculated betweenness attack. At each iteration, we study how the connected components change as more nodes are eliminated. To keep track of the eliminated nodes we use

$$f = \frac{\text{Total number of removed vertices up to now}}{N} = dq \times (\text{round number})$$

while, for the connected components, we follow two quantities:

$$S = \frac{N_{GCC}}{N_{surv}} = \frac{N_{GCC}}{(1-f)N}$$

$$\langle s \rangle = \frac{\sum(\text{sizes of non-GCC components})}{(\text{number of non-GCC components})}$$

where N_{GCC} is the size of the GCC, N_{surv} the size of the full network at the step taken in consideration and N is the initial network's size.

The measure S is informative of the state of the connectivity of the survived nodes while $\langle s \rangle$ gives information on the average non-GCC components sizes.

Random Attack. In the random attack, the nodes removed at each step are chosen randomly. This strategy models accidental failures or malfunctions that affect nodes independently of their type or role in the network. It therefore measures the robustness of the network under non-targeted attacks, since all nodes have the same probability of being removed.

Recalculated Highest-Degree Attack. The highest-degree attack targets the most connected nodes in the network. At each iteration, the nodes with the largest degree are removed. The term “recalculated” means that after each removal, all node degrees are recomputed on the updated network. This strategy is motivated by the fact that high-degree nodes often act as hubs and play a crucial role in network connectivity. Their removal is therefore expected to quickly fragment the graph.

Recalculated Betweenness Attack. The betweenness attack targets the nodes with the highest betweenness centrality, computed as in 2.2. As in the highest-degree attack, centrality values are recomputed after each removal step.

Rather than focusing on hubs, this strategy identifies nodes that act as bridges between different regions of the network. Removing such nodes can disconnect communities and therefore still rapidly fragment the graph.

2.4 Synthetic Network Models

In this report we generate synthetic networks using the algorithms provided by the `igraph` library. Unless otherwise specified, all networks are undirected. Most models allow the option of generating directed graphs, with the exception of the Watts–Strogatz model, which can only be generated as an undirected graph.

2.4.1 Erdős–Rényi (ER)

The Erdős–Rényi model is generated using:

```
igraph.GraphBase.Erdos_Renyi(n,p),
```

where n is the number of nodes and p is the probability that an edge exists between any pair of nodes. This construction produces a random graph with a binomial (approximately Poisson for large n) degree distribution.

2.4.2 Barabási–Albert (BA)

The Barabási–Albert model is generated using:

```
igraph.GraphBase.Barabasi(n,m),
```

where **n** is the total number of nodes and **m** is the number of edges created by each newly added node. The algorithm implements *preferential attachment*: starting from an initial fully connected core of **m** nodes, each new node attaches to **m** existing nodes. The probability of choosing an existing node **i** is proportional to its degree:

$$P(i) = \frac{k_i}{\sum_j k_j}.$$

2.4.3 Stochastic Block Model (SBM)

The Stochastic Block Model is generated using:

```
igraph.GraphBase.SBM(n,pref_matrix, block_sizes)
```

where **n** is the number of nodes, **block_sizes** specifies the size of each community, and **pref_matrix** is the matrix of connection probabilities between and within communities.

High modularity is obtained when intra-community probabilities are significantly larger than inter-community probabilities. Within each block, the model behaves like an Erdős–Rényi graph with its own connection probability.

2.4.4 Watts–Strogatz (WS)

The Watts–Strogatz model generates a small-world network by starting from a regular lattice and rewiring edges with probability **p**. The model is created using:

```
igraph.GraphBase.Watts_Strogatz(dim,size,nei,p)
```

where **dim** is the number of lattice dimensions, **size** is the number of nodes along each dimension, **nei** determines the initial neighborhood radius in the lattice, and **p** is the rewiring probability.

For **p=0**, the network is a regular lattice with high clustering and long average path length. As **p** increases, random long-range edges are introduced, reducing the average path length while preserving clustering, thus producing the characteristic small-world structure.

In our experiments we use **dim = 1** (a ring lattice) and **nei = 2**, meaning that each node is initially connected to its four nearest neighbors before rewiring.

3 Real-World Networks: Characterisation and Robustness

In this section we apply the characterisation described in 2.1 and 2.2 to the real-world networks and give an interpretation of such results. Then we apply the attack strategies as in 2.3 and report the behaviours of S and $\langle s \rangle$, focusing on their degree distribution and on its relationship with the betweenness distribution. This analysis provides insight into how each network is organised and highlights the structural patterns that will be discussed in the following subsections.

3.1 Power-Grid dataset

Datasets. In this section we study the European Power-Grid (EU) and the United States Power-Grid (US) networks. The nodes represent stations (e.g. generating stations) and edges represent high voltage lines. The data are taken from [10] and [8].

We observe from our general analysis (reported below), low degree averages which might be explained by the fact that the Power-Grid networks are strongly embedded in geographic space. As a consequence, a vertex can not be connected to any other vertex inside the network, but only to the ones that are feasible from an infrastructure perspective. Hence the degree distribution is not expected to

display pronounced heavy tails and is expected to have a relatively low average degree. Both graphs present a strong modularity which suggests a strong division between well determined communities. The interpretation of such communities is not obvious as they may represent, for example, densely inhabited that require high electrical power or geographically isolated regions with few connections to the rest of the network (e.g. mountains, sea) or politically determined regions. To assess which interpretation is beyond the scope of this project and would require additional information.

Metric	Europe	US
Number of nodes (N)	13844	4941
Number of edges (L)	17276	6593
Density	0.00018	0.00054
Average degree (k_{avg})	2.50	2.67
Degree variance ($\text{Var}(k)$)	1.66	3.21
Clustering coefficient	0.1126	0.1065
GCC size	13478	4941
Average shortest path	49.51	18.99
Assortativity	0.0146	0.0036
Number of communities	78	40
Modularity	0.9663	0.9377

Table 1: Structural comparison between the European and US Power Grid networks

By looking at the Degree Distribution $P(k)$ and at its Complementary Cumulative Distribution (CCDF) we observe the predicted absence of heavy tails as well as a peak close to the average degree. The observed distributions do not appear compatible with a power-law tail. An exponential tail appears more plausible, although no formal distribution fitting was performed. Following the arguments of [4], the absence of heavy-tailed degree distributions suggests a lower resilience to random failures than that observed in scale-free networks.

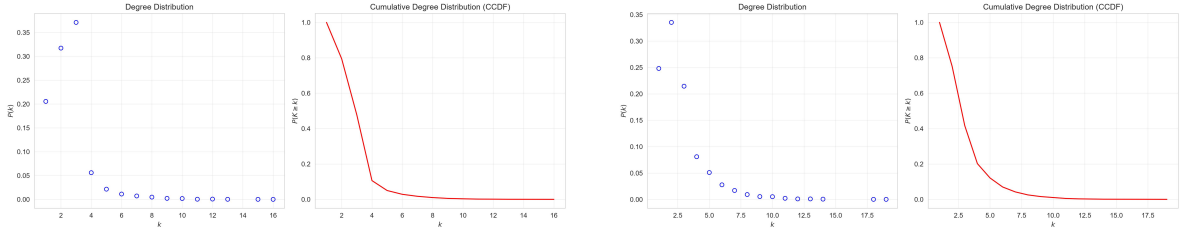


Figure 1: Comparison of degree distributions and CCDF between Europe (left) and the United States (right).

In literature one can find different fitting proposals for $P(k)$ of Power Grids (as for example in [11] which sums a geometric distributed random variable with an irregular discrete random variable) but we limit ourselves to observe an exponential decay for high values of k and a deviation from this behaviour for low k .

Here we look at the correlations between three centrality measures: Degree Centrality, Betweenness and PageRank. The information obtained is useful for two objectives: better understanding the structure of the network and assessing whether or not a highest-betweenness attack gives significantly different results with respect to the random and highest- k attacks.

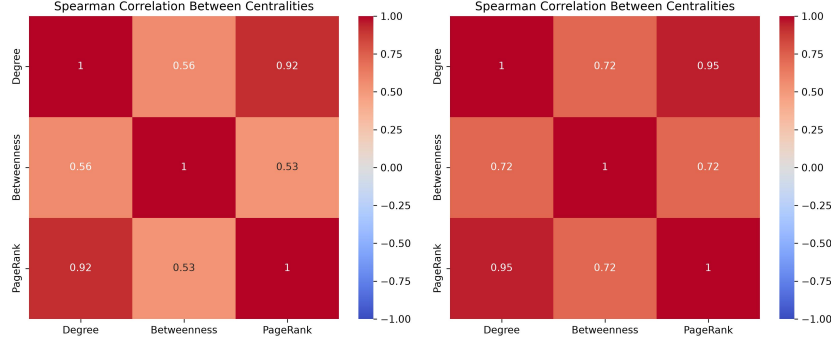


Figure 2: Comparison of the Spearman correlation among degree, betweenness, and PageRank centrality in the two power grids: Europe (left) and the United States (right).

The very high Spearman correlation between PageRank and Degree ($\rho_{EU} \sim 0.92$, US: $\rho_{US} \sim 0.95$) indicates that the two metrics produce highly similar node rankings. Hence we can say that, in these networks, PageRank and Degree identify largely overlapping sets of important nodes. On the other hand, the correlation between PageRank and Betweenness ($\rho_{EU} \sim 0.53$, $\rho_{US} \sim 0.72$) tells us that bridge nodes can partially overlap with highly connected and high-PageRank nodes (hence well connected nodes inside communities), but there are many cases where such bridges have low k and low PageRank. These nodes may act as bottlenecks between communities. We conclude that we expect a meaningful difference in the robustness behaviour between highest- k and highest-betweenness attacks.

3.1.1 Robustness Analysis

In this section we take a look at the response of the Power-Grid networks under three types of attack described in 2.3 and we would like to interpret these attacks as failures of power-grid's stations. However, since we do not take into account physical effects of such failures (as, for example, load redistribution and the possible consequent avalanche effects) the interpretation offered is not satisfying. Hence we look at the results as more related to the topology of the network rather than the real-world properties of its nodes.

We observe that the highest- k attack is more effective at disrupting the Giant Connected Component (GCC) than a random attack. This result is consistent with the correlation patterns observed among centrality measures. In particular, the strong correlation between Degree and PageRank indicates that targeting hub nodes also tends to remove nodes with high PageRank, namely nodes that occupy highly influential positions within the network and contribute significantly to local connectivity. Consequently, degree-based attacks preferentially target highly connected nodes that are likely to play a major role in maintaining connectivity within communities.

Moreover, the moderate correlation between Degree and Betweenness Centrality suggests a partial overlap between hubs and nodes that mediate communication among communities. As a consequence, some of the targeted high-degree nodes may also play a bridging role, contributing to inter-community connectivity.

The combined disruption of both highly connected nodes and, in some cases, key bridging nodes makes targeted highest- k attacks substantially more effective than random attacks in fragmenting the network and reducing the size of the GCC.

Looking at the response to the highest-betweenness attack we observe an higher disruptive effect on the GCC of such attack with respect to the other ones. This aligns with what we predicted before. It is particularly interesting how the highest-betweenness attack causes a pronounced increase in $\langle s \rangle$. We interpret this as a consequence of removing nodes that likely act as bridges between different communities leaving their internal structure largely unchanged. As a result, the GCC is rapidly broken into relatively large components, which may approximately correspond to the original community structure. In contrast, the highest- k and random attacks do affect the internal structure of such communities. Consequently, the resulting components tend to be smaller than the original communities yielding much smaller values of $\langle s \rangle$.

In conclusion, the extremely high modularity of both networks suggests that connectivity between

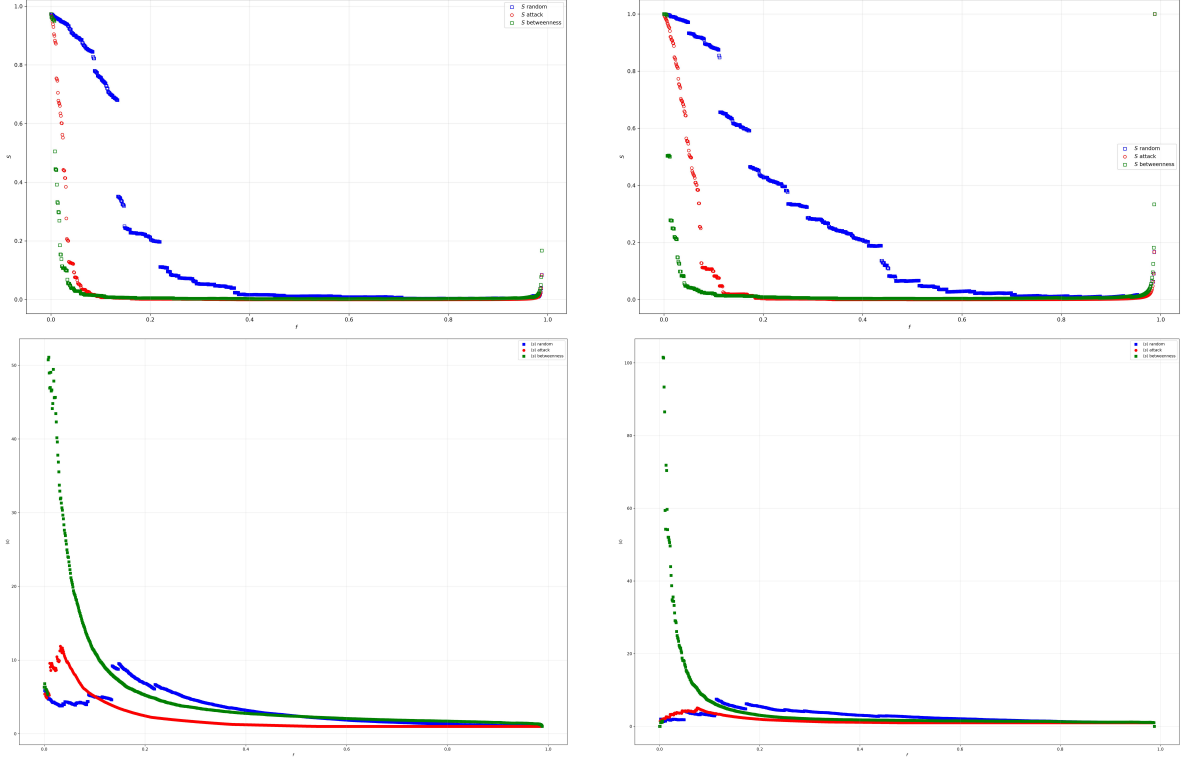


Figure 3: Behaviour of S and $\langle s \rangle$ of EU Power-Grid (left) and US Power-Grid (right) under the three kinds of attacks.

communities relies on a relatively small number of inter-community links. Since betweenness centrality is particularly effective at identifying the nodes supporting these connections, the superior performance of the adaptive betweenness attack is consistent with the strong community structure observed during the characterization phase.

3.2 Transport networks

Datasets. In this section, we consider two transportation networks. The first is a Flight Network, represented as a directed graph, where nodes correspond to airports and edges represent flight routes. The second is a Road Network, represented as an undirected graph, where nodes correspond to road intersections and edges to road segments. The data are taken from [9] and [3].

The global network properties reported in the table show that both networks are sparse, although the Road Network is considerably sparser. It also exhibits a much lower average degree, reflecting the strong geographical constraints that limit the number of connections that can be established at each intersection. Such constraints are substantially weaker in the Flight Network, where airports can maintain direct connections over very large distances, resulting in a higher average degree. These structural differences are also expected to affect path lengths. Flight routes create long-range shortcuts that reduce the average shortest path, whereas road networks rely on geographically local connections and therefore tend to exhibit longer paths between nodes. Overall, the strong spatial embedding of the Road Network makes its structure markedly different from that of the Flight Network and, in several respects, more similar to infrastructure networks such as power grids.

Besides, looking at the modularity we observe an extremely high value for the Road Network (corroborating the structural similarities observed with the Power-Grid networks) indicating a structure characterized by well-defined communities with, in comparison, sparse inter-community connectivity. On the other hand, the moderately high modularity of the Flights Network can be interpreted as a sign of a clear subdivision in communities which, nonetheless, are well connected between each other.

Metric	Flights	Roads
Number of nodes (N)	2939	1965206
Number of edges (L)	30500	2766607
Density	0.00353	1.43×10^{-6}
Average degree (k_{avg})	10.38 (in/out)	2.82
Degree variance ($\text{Var}(k)$)	466 (in), 469 (out)	0.99
Clustering coefficient	0.5889	0.0555
GCC size	2905 (weak)	1957027
Average shortest path	4.13 (strong GCC)	310.59
Assortativity	0.0459	0.1260
Number of communities	21	377
Modularity	0.6493	0.9926

Table 2: Structural comparison between the Flights and Road networks

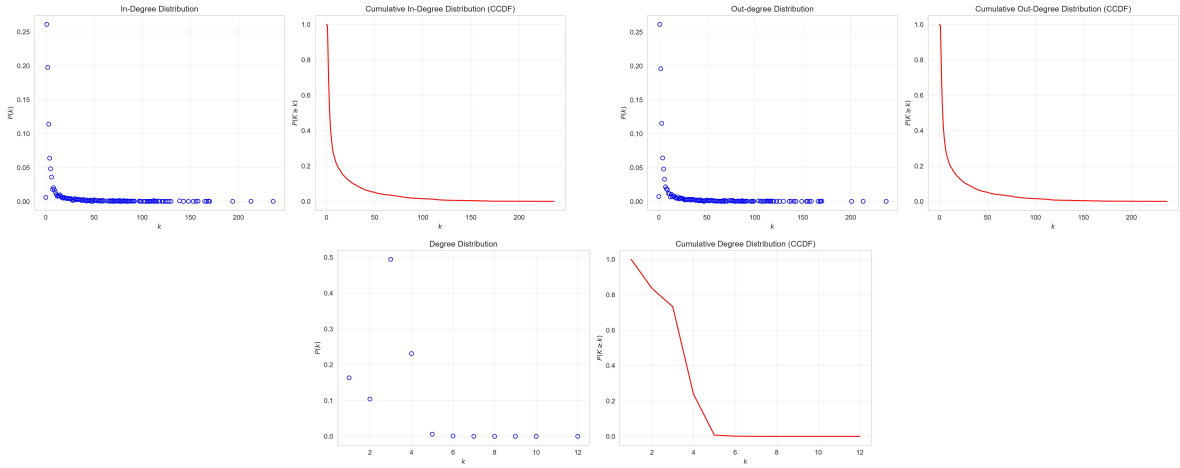


Figure 4: In-degree and out-degree distributions and CCDFs of the Flights network (top) and degree distribution and CCDF of Roads network (bottom)

As one expects for a flight network, the behaviour of in-degree and out-degree distribution are very similar since an airport A with an incoming flight from an airport B will probably have an outgoing flight from A to B. Also, since it is common to have important airports attracting most incoming and outgoing flights, it is not surprising to have seemingly power-law degree distributions. On the other side, the roads network topology is strongly influenced by its embedding in its geometric space which does not allow a node to connect to any other. Also, due to the typical structure of a crossroad, high degree nodes are absent and the distribution has most of its weight on low degrees ($\sim 3 - 4$).

The centralities correlations of the Flights Network confirm the high overlap of the in and out degree distributions in

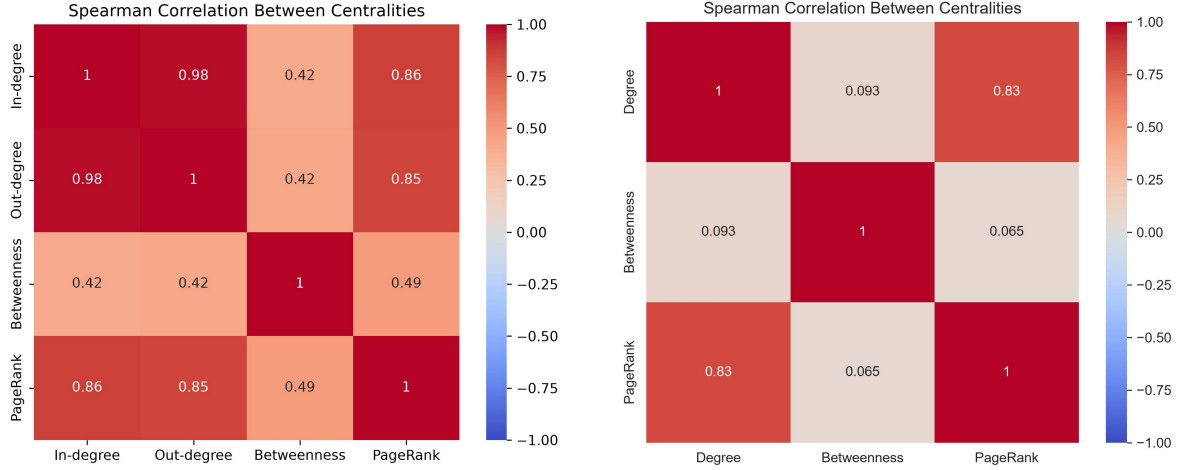


Figure 5: Comparison of the Spearman correlation among degree, betweenness, and PageRank centrality in the two transportation networks: Flights (left) and the Roads (right).

The nearly perfect correlation between in-degree and out-degree is consistent with the bidirectional nature of most flight routes. The strong correlations between Degree and PageRank indicate that highly connected airports usually coincide with the high-PageRank nodes. In contrast, the moderate correlations involving Betweenness Centrality suggest that even though there is a tendency of influential nodes to be also bridge nodes, still there is a considerable portion of bridge nodes that are not influential. Thus, while some major airports simultaneously act as regional hubs and inter-community connectors, many bridging airports derive their importance from their strategic location rather than from an exceptionally high number of connections.

The combination of high modularity, strong Degree-PageRank correlation, and negligible correlations involving Betweenness Centrality suggests a clear separation between nodes responsible for local cohesion and nodes responsible for inter-community connectivity. High-degree and high-PageRank nodes appear to play a role distinct from the nodes identified by Betweenness Centrality. Whereas the potential inter-community bridges are primarily identified by Betweenness Centrality. This is consistent with a strongly modular network composed of densely connected communities linked by a relatively small number of bridging nodes.

Given the stronger community structure of the Road Network, Betweenness Centrality identifies a small set of nodes that are likely to be crucial for maintaining connectivity between communities that would otherwise be weakly connected. As a consequence, a highest-betweenness attack is expected to be significantly more effective than a highest-degree attack. In the Flight Network, where communities are less isolated and hubs often contribute to both local and global connectivity, the aforementioned effectiveness of betweenness-based attacks is expected to be still valid but to a much smaller extent.

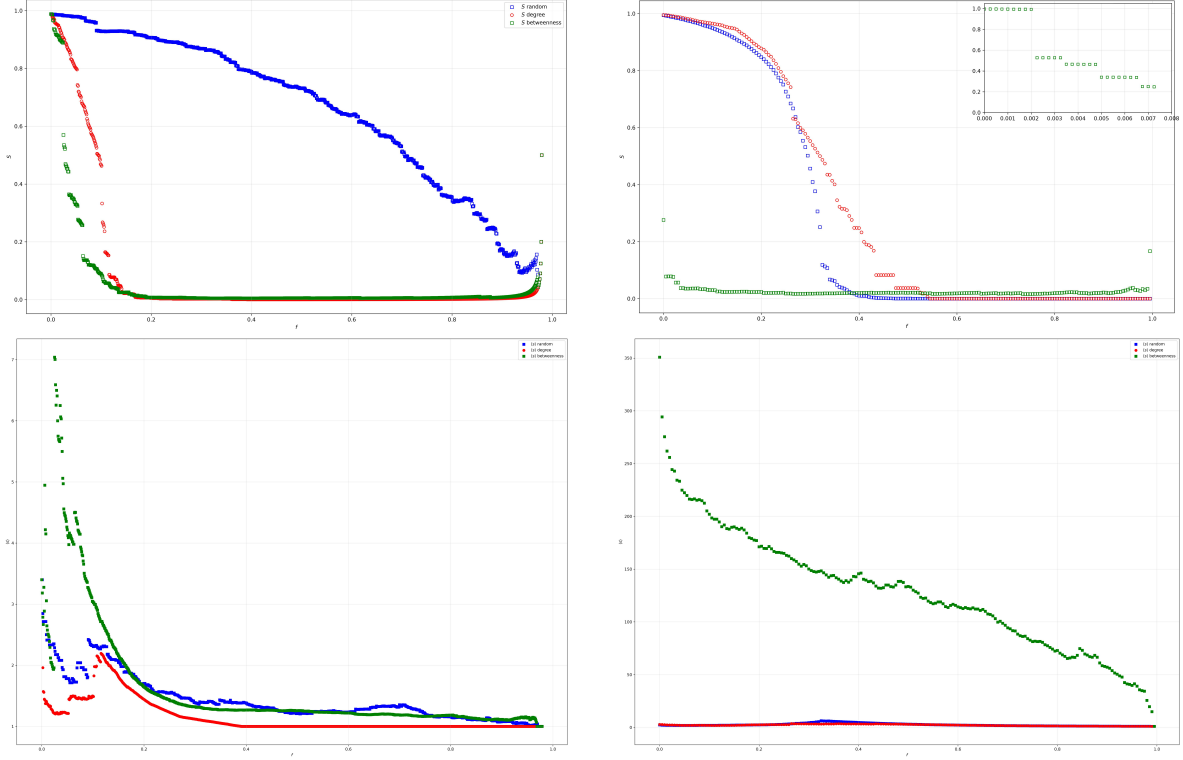


Figure 6: Behaviour of S and $\langle s \rangle$ of Flights (left) and Roads (right) under the three kinds of attacks.

The observed behaviour is in line with what we expected. In both cases the betweenness attack is the most effective strategy, which is consistent with the modular structure observed during the characterization phase. The effectiveness difference is much higher on the Roads network, which may be related both to its extremely high modularity and to the weak overlap between Degree and Betweenness rankings observed in Figure 5. The inset shows that most of the GCC is destroyed during the first few iterations, confirming the effectiveness of such an attack.

Also, the Roads show an interesting behaviour: the random attack appears to be more effective than the highest- k one. A possible explanation is the extremely narrow degree distribution of the Road Network. Since most nodes have very similar degrees, targeting the highest-degree nodes provides only limited additional information relative to random selection.

Besides we observe that $\langle s \rangle$ under the betweenness attack has a much larger peak than the one under random and highest- k attacks. This can be explained by the same mechanism as in the Power-Grid case. Again, the different level of modularity plays a role: while the Flights network shows a difference between the peaks of $\langle s \rangle$ of the same order as the peaks values, we get a peak difference in the Roads case two orders of magnitude.

In conclusion we observe a high robustness of the Flights network under random failures analogous to the scale-free networks. Roads, on the other hand, show a behaviour resembling a percolation transition under random attacks.

3.3 Social networks and Web Graph

Datasets. In this section, we consider three social networks: the Facebook Artists (FbA), the Facebook New Sites (FbN) and the Web Google (WG). In the two Facebook networks the nodes represent official Facebook pages while the links are mutual likes between sites, making it an undirected network. In WG the nodes represent web pages and directed edges represent hyperlinks between them. WG is then a directed graph. The datasets are taken from [2, 1].

The global network properties reported in the table show that all three networks are sparse, but WG is two orders of magnitude less dense than the others. WG also exhibits a lower average degree and very strong modularity. FbA and FbN on the other hand, both have a modularity $Q \sim 0.6$ indicating a clear

community organization (even if less clear pronounced than WG). Degree assortativity is negligible in all the graphs.

Metric	Fb Artists	Fb New Sites	Web Google
Number of nodes (N)	50515	27917	875713
Number of edges (L)	819090	205964	5105038
Density	0.00064	0.00053	6.66×10^{-6}
Average degree (k_{avg})	32.43	14.76	5.83 (in/out)
Degree variance ($\text{Var}(k)$)	4029	528	1539 (in), 43 (out)
Clustering coefficient	0.1472	0.3199	0.6223
GCC size	50515	27917	855802 (weak)
Average shortest path	3.69	4.39	11.53 (strong GCC)
Assortativity	-0.0191	0.0219	-0.0652
Number of communities	35	45	644
Modularity	0.6152	0.6331	0.9796

Table 3: Structural comparison between the Facebook Artists and News Sites (undirected) and Web Google (directed) networks

The weak GCC treats the network as undirected, ignoring link direction. The strong GCC, instead, requires that nodes can reach each other respecting the actual direction of the links.

We observe that the $P(k)$ s and CCDFs in loglog scale show linear behaviour in intervals of the k domain. This indicates that, in such regions, the behaviour might be power-law.

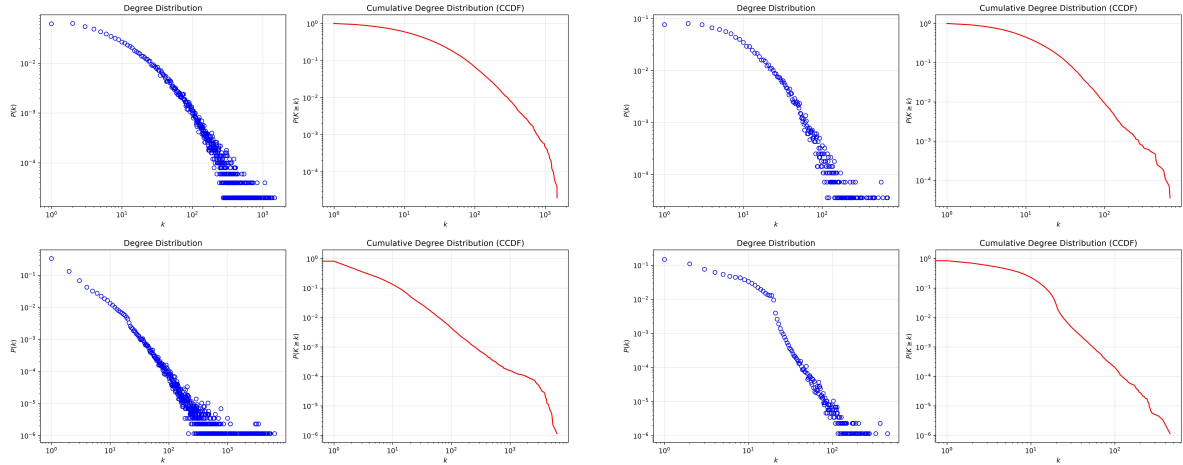


Figure 7: Degree distribution and CCDF of FbA (top left), FbN (top right) networks and in-degree (bottom left) and out-degree (bottom right) distributions and CCDFs of the WG network.

Looking at the correlations between centralities of the two Facebook we observe that Degree and PageRank produce highly similar node rankings, suggesting that PageRank is strongly associated with local connectivity in these networks. In contrast, Betweenness centrality captures a different aspect of node importance. The weak correlations between Betweenness and both Degree and PageRank suggest that bridging roles are only partially associated with popularity. The effect is especially strong in FbA, where bridge nodes appear largely distinct from highly connected nodes despite the presence of strong community structure. Therefore, we expected betweenness-based attacks to be particularly effective, since the nodes identified by Betweenness Centrality appear structurally distinct from the highly connected nodes.

In the WG network, instead, we observe that the In-degree Centrality is highly correlated with the PageRank while the Out-degree shows little correlation with PageRank. This behaviour is consistent with the interpretation of PageRank as a measure that rewards nodes receiving links from other important nodes.: it gives high authority of web pages that are linked many times by other pages, whereas a high out-degree alone does not necessarily imply a high PageRank score. This is expected

for this kind of graphs, since it is reasonable to expect that highly cited pages occupy structurally important positions within the hyperlink network. We also observe that, due to the non-null in/out degree correlation, pages receiving many links have also a mild tendency to create many links. At the same time the distinction between pages that primarily receive links and pages that primarily create links appears evident. We also observe that nodes with high PageRank exhibit a moderate tendency to also have high Betweenness Centrality. A similar, although weaker, tendency is observed for high out-degree nodes.

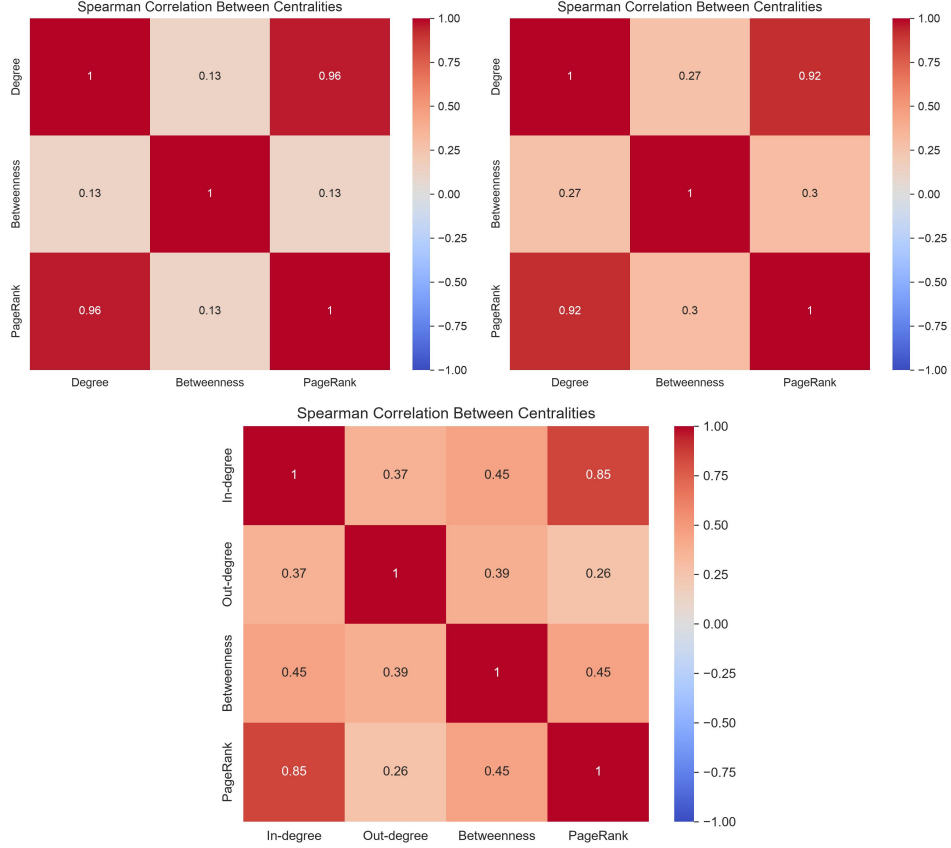


Figure 8: Comparison of the Spearman correlation among degree, betweenness, and PageRank centrality for the FbA, FbN (top) and the WG (bottom) networks.

The previous picture of the Facebook networks has to be changed once the attacks are performed. Indeed the robustness profile of FbA shows that the highest-degree nodes causes a slightly faster collapse of the giant connected component than removal based on betweenness, suggesting that high- k nodes play a slightly more important role than bridging nodes in maintaining global connectivity. On the other hand, for FbN betweenness-based attacks are marginally more effective, suggesting greater reliance on bridging nodes linking different regions of the network. The attack results suggest then, that both hub nodes and bridging nodes contribute to the overall connectivity of the Facebook networks, without either structure clearly dominating the other. In other words: the two networks seem to be structurally cohesive due to both hub nodes and bridge nodes, but neither of these node types dominate the networks topologies. We also observe a strong resilience against random failures, typical of scale-free networks.

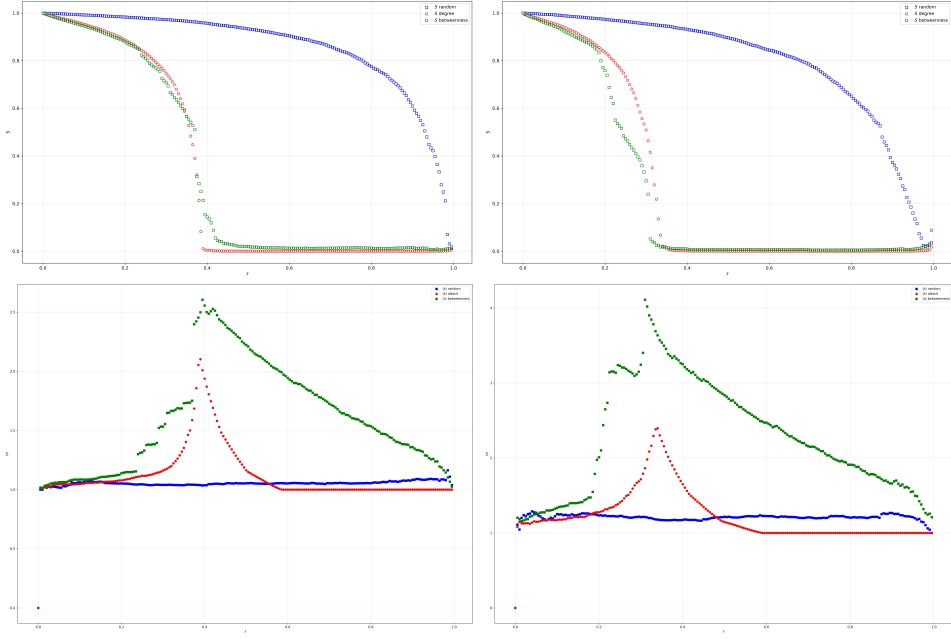


Figure 9: Behaviour of S and $\langle s \rangle$ of FbA (top left) and FbN (right) under the three kinds of attacks.

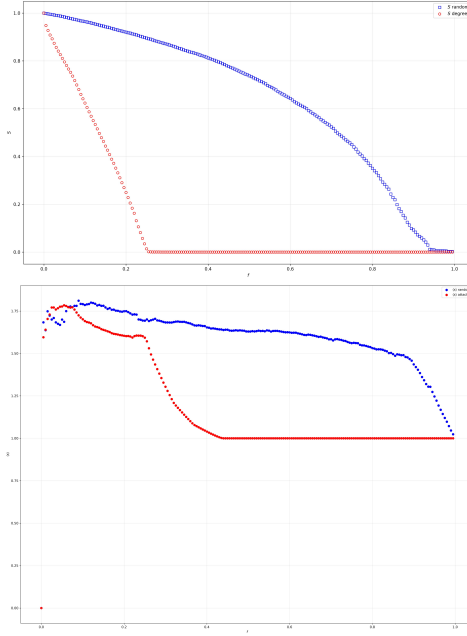


Figure 10: Behaviour of S and $\langle s \rangle$ of FbA (top left) and FbN (right) under the three kinds of attacks.

The Web Google network contains $N = 875,713$ nodes and $L = 5,105,038$ edges. Even using the approximate betweenness centrality algorithm, the adaptive attack would require hundreds of successive recomputations of a global shortest-path-based metric. As a result, the total computational cost becomes prohibitive for networks of this scale on the hardware at our disposal. In contrast the degree centrality can be updated efficiently from local information and does not depend on the global structure of the graph. The computational cost of this attack is then manageable.

Under highest- k and the random attacks we observe that the former is extremely effective, as one expects from a network displaying a highly heterogeneous degree distribution. On the other hand the network proves to be very robust under latter attack.

4 Synthetic networks

In this section we analyse the four synthetic networks generated as described in 2.4 starting from the global characterisation. The Erdős-Rényi network exhibits homogeneous random connectivity with negligible clustering and no prominent hubs. The Barabási-Albert network is strongly centralized around a small number of highly connected hubs, producing short path lengths and weak community organization. The Watts-Strogatz network displays highly clustered local neighborhoods. Interestingly, community detection yields an extremely high modularity. This likely reflects the near one-dimensional ring structure of the network, which can be partitioned into densely internally connected regions linked by relatively few rewired shortcuts. Finally, the Stochastic Block Model is characterized by pronounced mesoscopic organization, with numerous well-defined communities connected through sparse inter-community links, making community structure the primary determinant of its topology.

Metric	Erdős-Rényi	Barabási-Albert	Watts-Strogatz	SBM
Number of nodes (N)	10000	10000	5000	100000
Number of edges (L)	13952	49985	10000	140632
Density	0.0002791	0.0009998	0.00080016	2.81×10^{-5}
Average degree (k_{avg})	2.7904	9.997	4.0	2.81
Degree variance ($\text{Var}(k)$)	2.7841	138.0846	0.0808	2.79
Clustering coefficient	0.0001350	0.0052695	0.4718	0.0018
GCC size	9274	10000	5000	92636
Average shortest path	8.8917	3.7331	25.99	12.34
Assortativity	-0.00290	-0.00395	-0.0223	-0.0015
Number of communities	53	15	69	95
Modularity	0.6912	0.2983	0.9519	0.9106

Table 4: Structural comparison between Erdős-Rényi, Barabási-Albert, Watts-Strogatz, and SBM networks

The observed degree distributions are consistent with the theoretical expectations of the corresponding models: approximately Poisson for ER and SBM, power-law-like for BA, and sharply concentrated around the average degree for WS. According to [5], scale-free networks with degree exponent $\gamma \leq 3$ are expected not to exhibit a finite percolation threshold in the thermodynamic limit. Hence we expect from BA that S will go to zero only due to finite size effects.

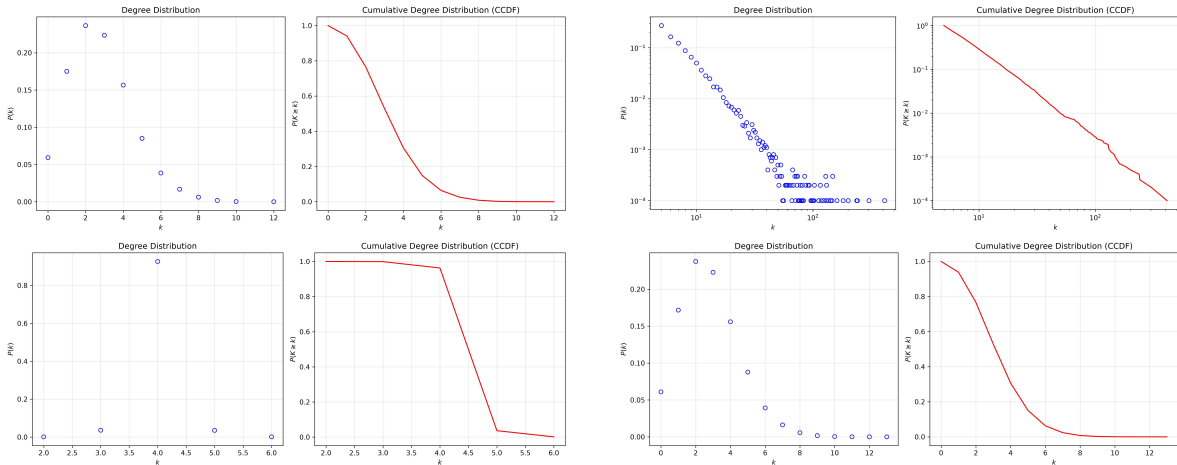


Figure 11: Degree distribution and CCDF of ER (top left), BA (top right in log-log scale), WS (bottom left) and SBM (bottom right) networks.

Looking at the correlations, we observe that Degree and PageRank produce highly similar rankings in all networks except the Watts-Strogatz model. This can be explained by the extremely narrow degree distribution of the latter ($\langle k \rangle = 4$, $\text{Var}(k) = 0.08$), which causes degree not to distinguish between

nodes. Since most nodes have nearly identical degree, PageRank becomes less dependent on the number of connections and more sensitive to the position of a node within the network. In particular, nodes located at the endpoints of rewired shortcut links or occupying structurally advantageous positions may accumulate noticeably higher PageRank scores despite having a degree similar to that of most other nodes. The very high modularity of the network further suggests that these positional effects can become important. Consequently, the Degree-PageRank moderate correlation ($\rho \approx 0.45$) indicates that PageRank captures aspects of node position that are not fully described by degree. Besides, the correlation between Degree and Betweenness is pretty low except for the Erdős-Rényi. Hence, in all the other networks the roles of bridge and hub are well distinguishable in most cases. In the ER network instead, high-degree nodes frequently coincide with high-betweenness nodes. This overlap of structural roles is consistent with the random organization of the network, where connectivity is relatively homogeneous and shortest paths are likely to traverse highly connected nodes. The Betweenness-PageRank correlation instead, changes for all the networks. In ER, due to the lack of a clear community structure, we observe a high correlation indicating the most of the relevant nodes are also traversed by many shortest paths. In the BA network we observe that the roles of hub and bridge are only partially overlapping, while on the SBM the low correlation matches with the extremely high modularity of the network. In the latter case, we can interpret the hubs as being relevant nodes inside well defined communities while the bridging nodes as relevant nodes for the inter-community communication. Finally, in the WS network the two metrics are completely uncorrelated since PageRank is driven mainly by position relative to the few rewired shortcuts while Betweenness is driven by whether a node lies on one of those shortcuts. This because a shortcut endpoint may lie on a disproportionately large number of shortest paths causing the Betweenness to increase, while PageRank may not increase much because the node receives no extra links.

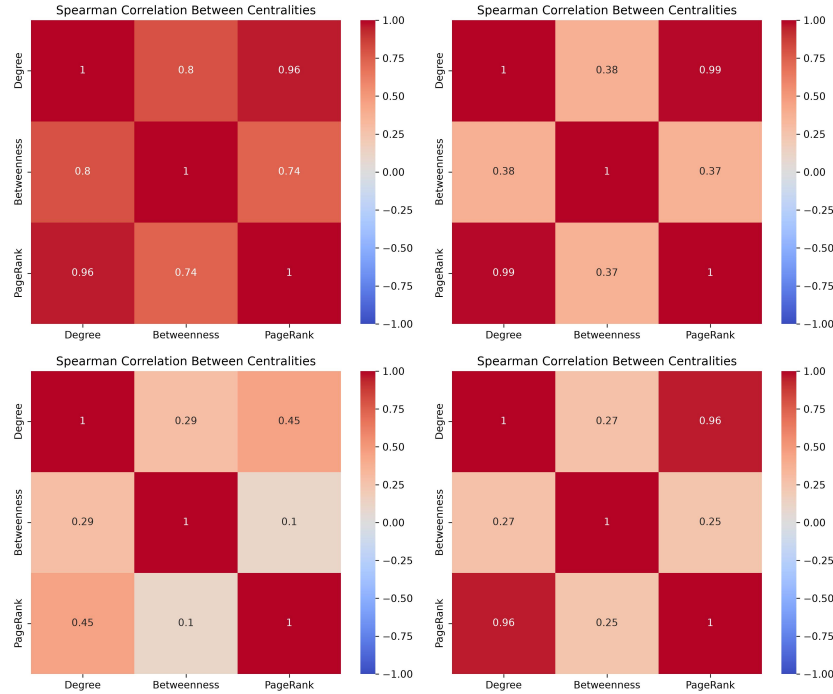


Figure 12: Comparison of the Spearman correlation among degree, betweenness, and PageRank centralities the ER (top left), BA (top right), WS (bottom left) and SBM (bottom right) networks.

4.1 Robustness Analysis

The effectiveness of attack strategies is strongly dependent on the dominant structural feature of each network. In these realizations of the Erdős-Rényi and the Barabási-Albert networks, degree-based attacks initially produce the fastest destruction of the giant connected component (GCC), reflecting the importance of highly connected nodes. However, adaptive betweenness attacks eventually become more destructive, as betweenness identifies newly emerging critical bridging nodes after each removal.

Random attacks are consistently the least effective. In this realization of the Watts-Strogatz network, adaptive betweenness attacks clearly outperform all other strategies, suggesting that global connectivity relies on a small number of bridging nodes linking densely connected neighborhoods. Remarkably, degree-based attacks eventually become less effective than random failures during later stages of the attack, suggesting that node degree is a poor indicator of structural importance in this highly modular network. In this realization of the Stochastic Block Model, degree-based attacks remain the most effective throughout the process, showing that node degree is a more effective indicator of structural vulnerability than betweenness centrality in this particular realization.

The SBM result is particularly interesting because it differs from WS, Road and Power-Grid networks, despite their similarly high modularity. This indicates that community structure alone is insufficient to predict attack effectiveness. The placement of the node inside the network of high-degree and high-betweenness nodes also plays an important role. In this SBM realization, degree appears to be a better indicator of structural vulnerability than betweenness.

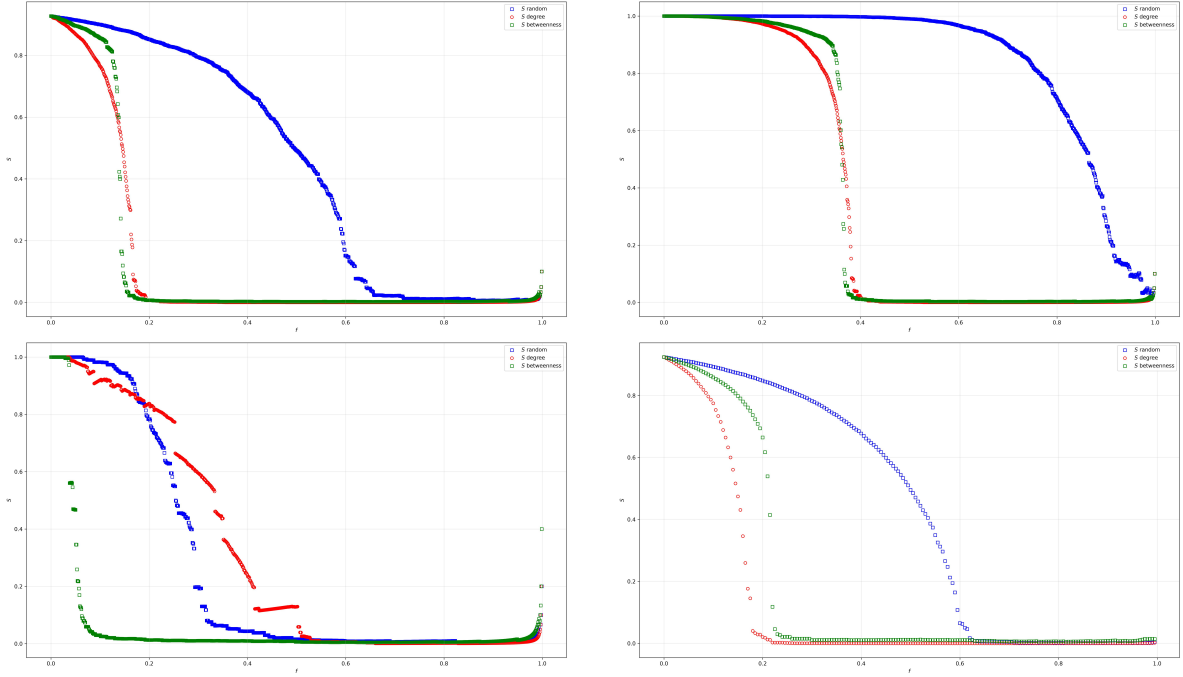


Figure 13: Behaviour of S of ER (top left), BA (top right), WS (bottom left) and SBM (bottom right) networks under the three kinds of attack.

The behaviour of $\langle s \rangle$ shows that betweenness based attack consistently breaks the GCC in larger components with respect to the random and degree based attacks. This is consistent with the removal of nodes connecting different regions of the network while leaving much of their internal structure intact. It is relevant to see how the WS network's $\langle s \rangle$ under betweenness attack presents a peak two orders of magnitude larger than the other types. This is consistent with the high effectiveness of such an attack at disrupting this network's GCC and confirms that WS has a sharp community structure.

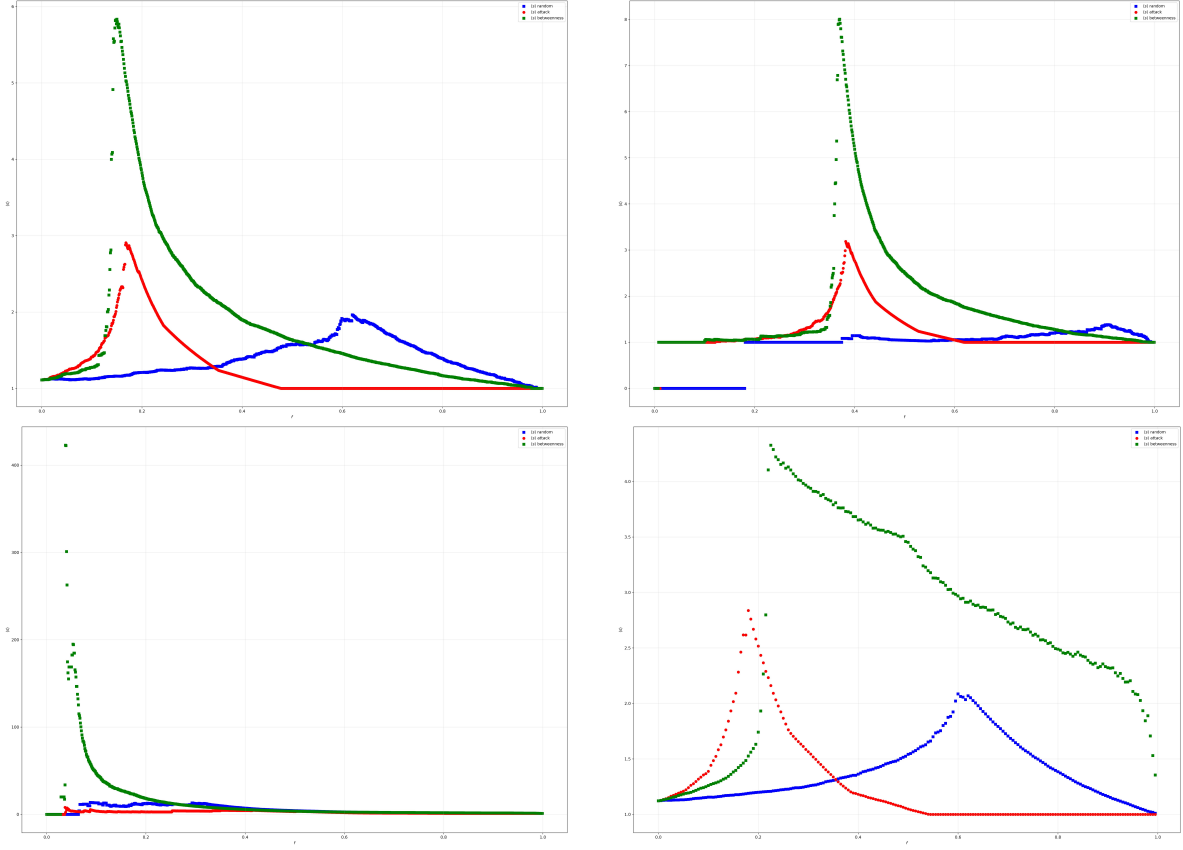


Figure 14: Behaviour of $\langle s \rangle$ of ER (top left), BA (top right), WS (bottom left) and SBM (bottom right) networks under the three kinds of attack.

5 Conclusion

In this project we investigated the robustness of a diverse set of real-world and synthetic networks against random failures and targeted attacks. Building on [4], we analyzed how structural properties (e.g. degree heterogeneity, community organization) influence the response of a network to node removal. We combined global topological characterization with robustness experiments based on random, recalculated highest-degree and recalculated betweenness-centrality attacks.

The results confirm that network topology plays a fundamental role in determining robustness. Networks displaying highly heterogeneous degree distributions, such as the Flight, Facebook, Web Google, and Barabási-Albert networks, exhibit a strong resilience against random failures while being significantly more vulnerable to targeted attacks on influential nodes. This is consistent with the classical picture of scale-free networks, where connectivity strongly relies on a relatively small number of highly connected nodes. Conversely, networks characterized by narrow degree distributions and strong spatial constraints, such as Power Grids and Road Networks, are generally less robust to random failures and do not necessarily exhibit a clear distinction between random and degree-based attacks.

A central objective of this work was to evaluate the effectiveness of adaptive betweenness-based attacks. In many cases, particularly in the Power-Grid, Road, and Watts-Strogatz networks, this strategy proved to be much more destructive than attacks based on degree centrality. These networks share a pronounced community structure, suggesting that global connectivity relies heavily on a relatively small set of nodes acting as bridges between densely connected regions. Removing such nodes rapidly fragments the giant connected component while leaving much of the internal community structure intact, hence producing large peaks in $\langle s \rangle$.

At the same time, our results show that high modularity alone is not sufficient to guarantee the superiority of betweenness-based attacks. The Stochastic Block Model provides a counterexample: despite its

strong community organization, degree-based attacks remained more effective throughout the removal process. This observation suggests that the relative importance of hubs and bridge nodes depends not only on the presence of communities but also on their arrangement in the network. As a consequence, predicting which is the most destructive attack strategy requires a more complete understanding of the interplay between degree distribution, community structure, and centrality correlations.

The correlation analysis between Degree, PageRank, and Betweenness Centrality further supports this interpretation. Networks where Degree and Betweenness rankings strongly overlap tend to exhibit similar responses to degree-based and betweenness-based attacks. In contrast, when the two measures identify distinct sets of nodes, adaptive betweenness attacks often become significantly more effective. Thus, centrality correlations provide valuable information about both the organization of the network and its potential vulnerabilities.

Overall, the results highlight that network robustness emerges from the combination of multiple structural features rather than from any single topological property. Degree heterogeneity remains a key factor in determining resilience to random failures, but community organization and bridge-node distribution can be equally important in determining vulnerability to targeted attacks.

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